

MARMUL, OmanWMO: **413040**Lat: **18.1411N**Lon: **55.1806E**Elev: **896**StdP: **14.23**Time Zone: **4.00 (E04)**Period: **98-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	51.1	54.0	15.7	12.6	76.3	20.8	16.1	75.9	26.9	75.5	24.0	73.8	7.0	220	0.481

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.9	111.2	68.4	109.6	68.7	108.0	68.9	77.4	85.7	76.7	85.8	75.8	86.9	10.8	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.0	140.3	80.2	74.7	134.5	79.9	73.5	128.7	79.6	41.8	85.4	41.0	84.9	40.1	86.2	81.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
26.5	23.2	20.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
			DB	36.1	78.1	2.2	1.3	34.5	79.1	33.2	79.9	32.0	80.6	30.3	81.6
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	82.2	67.2	70.5	77.4	84.2	90.1	94.3	95.0	93.1	87.3	81.8	75.5	69.7		
	DBStd	10.38	4.01	4.71	4.36	3.37	3.40	3.85	4.09	4.49	4.82	3.38	2.77	3.51		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	42	26	9	0	0	0	0	0	0	0	0	0	7		
	CDD50	11765	534	574	850	1026	1242	1329	1396	1336	1118	986	764	610		
	CDD65	6332	95	163	385	576	777	879	931	871	668	521	314	152		
	CDH74	85063	631	1310	3811	7004	11162	14035	15061	13483	8819	5772	2901	1074		
	CDH80	53302	63	345	1783	4083	7191	9866	10747	9327	5555	3104	1080	158		
(14) Wind	WSAvg	9.4	8.5	11.0	12.4	10.2	9.8	9.8	8.8	8.5	9.8	8.9	7.5	7.9		
(15) Precipitation	PrecAvg	3.7	0.3	0.3	0.5	0.4	0.2	0.2	0.2	0.2	0.1	0.2	0.4	0.5		
	PrecMax	7.8	0.7	0.9	3.5	3.3	1.8	2.4	1.0	1.5	0.7	1.4	4.2	1.7		
	PrecMin	1.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0		
	PrecStd	1.8	0.2	0.2	0.8	0.7	0.4	0.4	0.2	0.3	0.2	0.3	0.7	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	84.5	90.3	98.5	104.0	110.5	113.4	112.7	111.6	109.3	101.9	93.1	85.4		
		MCWB	61.1	64.5	64.2	66.0	67.2	67.2	68.7	70.0	68.7	65.2	63.5	64.8		
	2%	DB	81.7	87.2	94.9	101.3	107.4	111.3	111.0	109.6	106.3	98.9	90.2	83.4		
		MCWB	62.4	63.5	64.6	65.5	66.6	67.1	69.0	70.2	67.7	64.3	63.9	64.5		
	5%	DB	79.6	84.4	92.0	99.3	105.4	109.3	109.5	107.8	104.1	96.1	87.9	81.8		
		MCWB	62.9	63.0	63.9	65.3	66.6	67.2	69.5	70.7	67.7	64.2	65.0	64.5		
	10%	DB	77.7	81.7	89.0	96.7	103.4	107.1	107.6	105.7	101.2	93.5	85.9	80.1		
		MCWB	62.3	62.9	63.5	65.3	66.9	67.4	69.8	70.8	67.6	64.7	64.9	64.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.4	72.4	73.8	76.3	78.9	78.3	76.6	76.4	75.3	75.1	73.8	71.4		
		MCDB	77.6	80.2	81.9	83.7	85.5	87.1	96.2	95.3	87.8	80.6	79.7	78.5		
	2%	WB	67.9	70.6	72.7	74.9	77.6	77.5	75.4	75.3	74.1	74.0	72.4	69.6		
		MCDB	74.5	78.1	80.4	81.7	84.3	86.4	95.3	96.1	85.9	79.7	78.6	76.9		
	5%	WB	66.6	69.1	71.7	74.0	77.0	76.5	74.6	74.5	73.2	73.1	71.0	68.4		
		MCDB	73.7	76.4	79.7	81.8	83.8	86.1	95.1	95.5	84.7	79.2	78.2	76.0		
	10%	WB	65.3	67.8	70.7	73.0	76.2	75.3	73.8	73.7	72.4	72.2	69.7	67.3		
		MCDB	73.0	75.2	78.7	81.8	83.6	86.7	95.0	94.8	83.6	78.8	77.7	75.3		
(35) Mean Daily Temperature Range	MDBR		21.9	23.0	24.2	25.5	25.9	25.5	24.9	25.0	25.7	23.4	22.6	21.6		
	5% DB	MCDBR	22.4	24.9	27.5	28.3	28.4	27.1	25.9	25.6	28.2	27.0	24.1	22.3		
		MCWBR	8.9	9.1	9.8	11.2	10.8	8.9	6.2	5.8	7.9	11.2	9.4	8.9		
	5% WB	MCDBR	18.8	20.0	21.0	23.9	22.8	22.9	24.1	23.3	23.2	19.9	20.2	19.6		
		MCWBR	7.8	6.6	6.1	7.8	7.6	6.8	4.9	4.9	5.2	7.3	7.9	8.0		
(40) Clear-Sky Solar Irradiance	taub		0.398	0.423	0.466	0.504	0.574	0.723	0.773	0.705	0.561	0.430	0.416	0.403		
	taud		2.105	2.021	1.897	1.828	1.688	1.443	1.405	1.512	1.738	2.045	2.085	2.091		
	Ebn at Noon		278	277	269	259	239	203	194	209	240	271	270	272		
	Edh at Noon		48	55	64	69	78	99	103	93	74	52	48	47		
(44) All-Sky Solar Radiation	RadAvg		1687	1944	2168	2305	2361	2193	2044	2112	2124	2000	1722	1600		
	RadStd		38	35	71	61	50	94	96	54	37	44	52	44		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (4 neighbors)		+0.27	N/A	-2.74	N/A	N/A	-0.54	N/A	N/A	N/A	N/A	

Nomenclature: See separate page